



Project Report: Quiz Master - V1

Author:

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I am aspiring Data Scientist with a strong foundation in machine learning, and data analysis. I have recently completed a diploma in data science.

Description:

Quiz Master - V1 is a multi-user quiz management system designed to facilitate structured exam preparation. The project involves creating an admin dashboard for managing quizzes, subjects, chapters, and questions, while students can attempt quizzes and track their performance. The system ensures students can only access quizzes with upcoming deadlines.

Technologies Used:

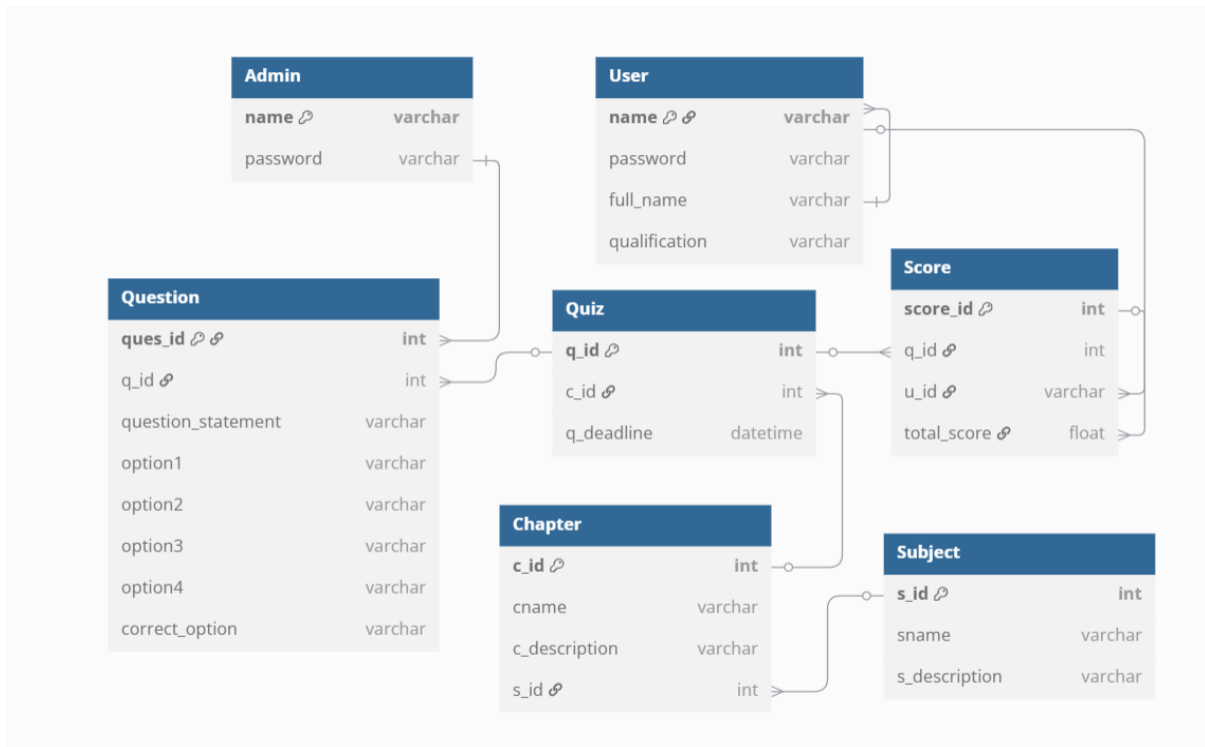
- **Backend:** Flask (Python), Flask-SQLAlchemy
- **Frontend:** Jinja2, HTML, CSS, Bootstrap
- **Database:** SQLite (programmatic creation)
- **Visualization:** Matplotlib
- **Purpose:** Flask is used for web development, SQLAlchemy for database, and Matplotlib for data visualization.

DB Schema Design:

The database consists of the following tables:

- **User:** (name, password, full_name, qualification) - Stores student information.
- **Admin:** (name, password) - Stores admin credentials.
- **Subject:** (s_id, sname, s_description) - Stores subjects.
- **Chapter:** (c_id, cname, c_description, s_id) - Stores chapters linked to subjects.
- **Quiz:** (q_id, c_id, q_deadline) - Stores quizzes linked to chapters.

- **Question:** (ques_id, q_id, question_statement, option1, option2, option3, option4, correct_option) - Stores quiz questions.
- **Score:** (score_id, q_id, u_id, total_score) - Stores student scores.



Reasons for Design Choices:

- Subjects and chapters are linked hierarchically to maintain course structure.
- Users and admins are stored separately for role management.
- Scores are stored separately for performance tracking.

Features:

User Authentication & Role Management

- Users and admins are stored in separate tables
- Admin has exclusive rights to manage data

Admin (Quiz Master) Features

- Manage subjects, chapters, quizzes, and questions
- Perform search operations on quizzes, and subjects
- View quiz and subject summary charts

User Features

- Register and log in
- View and attempt quizzes (only those with upcoming deadlines)
- View past performance through quiz summary charts

Search Functionality

- Admin can search for users, subjects, and quizzes
- Users can search for subjects and chapters

Performance Tracking

- Users can view quiz performance charts
- Admin can generate summary reports for subjects and quizzes

API Design:

- **Endpoints:**
 - /api/subjects (GET) - Fetch subjects.
 - /api/subject (POST) - Add subjects.
 - /api/edit_subject/s_id (PUT) - Update subject details.
 - /api/delete_subject/s_id (DELETE) - Delete a subject.
- **Implementation:** Flask-RESTful is used for API development.

Video link:

https://drive.google.com/file/d/1PkOByouNNUe_OliVxIM8dxJKgdBO1kPv/view?usp=sharing